

Evidence Before Authority

Technical architecture, safety controls and verified demonstration results

Release assurance for changes in agent authority

AgentGate compares a baseline and candidate synthetic agent release, identifies authority changes, checks controlled policy and assurance evidence, and returns an auditable release decision.

The release moment

For an engineering reviewer or release owner deciding whether an agent upgrade has enough evidence to receive additional authority.

The problem

An ordinary-looking upgrade can add order placement, write access or transaction authority while removing human approval. Passing stored tests does not make a policy violation acceptable.

CENTRAL DEMONSTRATION

Unsafe authority expansion: BLOCK

Two deterministic policy blockers prevent release approval, even when all ten selected stored assurance results pass.

Intended impact: make authority escalation visible and require accountable remediation before release. This demonstration uses synthetic manifests and stored evidence; it performs no trading.

Architecture

Three Foundry-backed model stages, constrained by authoritative Python



Teal = model reasoning. Blue-tinted = Python / application. This is a responsibility diagram; Python validates each handoff throughout the workflow.

Microsoft Foundry project-scoped Responses calls use the existing gpt-4.1-mini deployment. These are not persistent portal-agent resources. Explicit local deterministic mode remains available.

Governance and safety

Python owns authoritative decision fields; model explanations remain advisory

01 Exact comparison

Python validates manifests and computes the exact baseline-to-candidate diff. Model prose cannot add an authority change.

02 Controlled evidence

A bounded taxonomy, policy and test catalogue determine selected tests. Stored results are validated, not freshly executed candidate-agent tests.

03 Structured handoffs

Strict Pydantic contracts and reference checks constrain changes, plans and decisions. Each stage permits at most one structured-output repair.

04 Deterministic blockers

Python owns blockers, conditions, test selection and remediation. The model verdict must equal the deterministic floor in this implementation.

05 Safe failure

Unavailable Foundry calls, invalid output or invalid references stop the cloud review. Local mode is an explicit choice, never a silent replacement.

06 Human accountability

The reviewer inspects the evidence and required remediation. AgentGate issues an assurance decision; it does not deploy a release or place trades.

Model output cannot downgrade validated policy blockers.

Verified demonstration

Three live Foundry cases; nine stage calls accepted on their first attempt

UNSAFE

BLOCK

11.57 s model calls

SAFE

APPROVE

8.32 s model calls

INCOMPLETE

CONDITIONAL

7.98 s model calls

Live runs: 25 September 2026, 04:55 UTC. Durations sum model requests; they are not browser latency benchmarks. All three stages ran for each case.

Unsafe authority changes

Order-placement tool added | Brokerage READ to WRITE

Transaction authority: USD 0 to USD 5,000

Mandatory human approval removed

53

automated tests passed

13 / 13

repository evaluation cases passed

Evaluation = 10 labelled cases + 3 demonstrations. Six BLOCK-labelled cases reuse the unsafe manifest; these results do not establish general policy coverage or formal Foundry evaluation.

Evidence: docs/evidence/live-run-summary.json and scenario trace/decision pairs. Unsafe is blocked; safe is approved; missing evidence is conditional.

Evidence and reproduction

Inspect the record, reproduce the local review and understand the limits

Evidence that can be inspected

Local structured application JSON traces record stage status, Foundry response identifiers, timing, token usage and the final decision. The reviewer can download its review and trace. These are not Azure Monitor or Application Insights traces.

<https://github.com/Gabriel-vr1/AgentGate>

[Verified evidence index and detailed qualifications](#)

Windows / PowerShell: reproduce local mode

```
git clone https://github.com/Gabriel-vr1/AgentGate.git
cd AgentGate
py -3.12 -m venv .venv
.venv/Scripts/python.exe -m pip install -e '.[dev,foundry]'
.venv/Scripts/python.exe app.py serve
```

Open <http://127.0.0.1:8765>. Choose a scenario and Run review. For live mode, use an existing permitted Foundry project, Azure CLI sign-in, project endpoint and deployment settings described in README. No resources are provisioned by the application.

Limits and future work

Synthetic fixtures and a fixed risk mapping bound coverage. Model explanations remain advisory: schema validity does not establish narrative truth, and saved safe-case prose includes unsupported test commentary. Python-owned decision fields remain authoritative.

Persistent agent resources, portal telemetry, formal Foundry evaluation and fresh candidate-test execution are deferred. The reviewer is a local demonstration, without hosted authentication or production deployment. No commercial impact figures are claimed.

The human release owner remains accountable for the decision and remediation.